Monty Hall: Probability After a Reveal

Week 10 — a sample space, a host rule, and simulations

COMSC 2043 • Discrete Structures and Critical Thinking

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State the rules before calculating

- Three doors; one car, two goats.
- You choose door 1.
- The host knows the car location, opens an unchosen goat door, and chooses uniformly if both doors are goats.
- You may stay with door 1 or switch to the remaining closed door.

The nine equally likely cases

Each ordered pair (your pick, car location) has probability 1/9.

	Car 1	Car 2	Car 3
Pick 1	H opens 2 or 3	H opens 3	H opens 2
Pick 2	H opens 3	H opens 1 or 3	H opens 1
Pick 3	H opens 2	H opens 1	H opens 1 or 2

By symmetry, the three rows have identical stay/switch outcomes.

The three symmetric cases

Assume your pick is door 1.

Car location	Host opens	Switch outcome
1	2 or 3 (equally)	Lose
2	3	Win
3	2	Win

Stay wins in one of the three cases; switching wins in two.

Conditional probability is a ratio inside an event

For events A, B with $P(B) > 0$, definition: $P(A \mid B) = P(A \cap B)/P(B)$.

After a host reveal, the relevant sample space is the set of cases compatible with that reveal and the host's rule.

The denominator changes when we condition.

Bayes: host opens door 3

Let C_i mean car behind door i; let O_3 mean host opens door 3.

$P(C_1) = P(C_2) = P(C_3) = 1/3$.
$P(O_3 \mid C_1) = 1/2$, $P(O_3 \mid C_2) = 1$, $P(O_3 \mid C_3) = 0$.

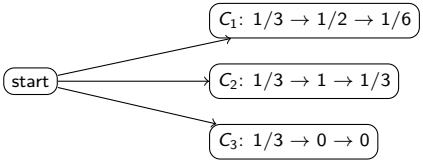
$P(O_3) = \sum_i P(O_3 \mid C_i)P(C_i) = 1/6 + 1/3 = 1/2$.

$P(C_1 \mid O_3) = ((1/2)(1/3))/(1/2) = 1/3$; hence $P(C_2 \mid O_3) = 2/3$. Switch.

Bayes denominator: add the paths that show door 3

Let O_3 mean the host opens door 3. The probability paths are: C_1 contributes $(1/3)(1/2) = 1/6$; C_2 contributes $(1/3)(1) = 1/3$; C_3 contributes $(1/3)(0) = 0$. Thus

$$P(O_3) = 1/6 + 1/3 + 0 = 1/2.$$



n doors: one closed alternative

You pick one of n doors. The first pick is correct with probability $1/n$.
A knowing host opens $n-2$ goats, leaving one alternative.
Switch wins exactly when the first pick was wrong: $P(\text{switch wins}) = (n-1)/n$.
For $n = 100$, this is $99/100$.

Open k goats; switch to one chosen survivor

There are $n-1$ goat-first-pick cases, each of probability $1/n$. The host opens k goats, leaving $n-1-k$ alternatives.

By symmetry, each specified survivor gets an equal share of the original wrong-pick probability:

$P(\text{win on a specified door}) = \frac{(n-1)/n}{n-1-k} = \frac{n-1}{n(n-1-k)}$.

Example $n = 5, k = 2$: each of two alternatives wins with probability $4/10$.

Monty Fall: a random host

Host chooses uniformly among the $n-1$ doors other than your pick, without knowing the car. A reveal is valid only if it shows a goat.
$P(\text{valid}) = P(C = \mathit{pick}) + P(C \neq \mathit{pick})P(\text{host avoids car} \mid C \neq \mathit{pick}) =$
$1/n + ((n-1)/n)((n-2)/(n-1)) = (n-1)/n$.
For three doors, $P(\text{host reveals car}) = 1/3$. Given a goat reveal, the two closed doors are equally likely: stay and switch each have probability $1/2$.

Monty Crawl: a leftmost-door rule

Three doors are numbered left to right. You pick door 1; the host opens the leftmost door that is neither your pick nor the car.
If car=1, host opens 2, so switch to 3 loses. If car=2, host opens 3, switch to 2 wins. If car=3, host opens 2, switch to 3 wins.
Thus unconditional switch win probability is $2/3$. But after observing door 2 open, the car is behind 1 or 3 with equal probability; after door 3 opens, car=2 with certainty.

Three Prisoners

A, B, C have equal chance $1/3$ of being pardoned; the guard knows and will name a different prisoner who will not be freed. A asks for a name.

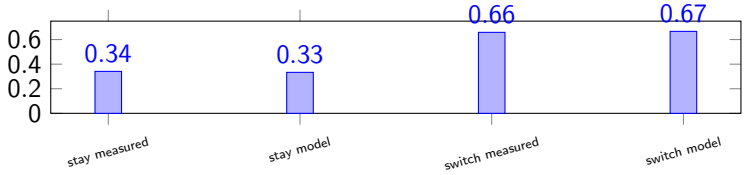
If A is pardoned (probability $1/3$), guard names B or C with equal chance. If B is pardoned, guard must name C; if C is pardoned, guard must name B.

Hearing "B will not be freed" leaves A's chance $1/3$ and raises C's chance to $2/3$. Naming B does not change A's chance.

Simulation checks a model's long-run rate

The simulator repeats the stated random experiment. For N independent trials, the law of large numbers says the observed fraction tends toward the model probability as N grows.

It does not prove the assumptions or promise a close result for a small N. Seeded measurement (10,000 games): stay $3412/10000 = 0.3412$; switch $6588/10000 = 0.6588$.



Playground: Buffon's hot dogs

A segment of length $\ell \le t$ crosses lines spaced t with probability $p = (1/\pi)\int_0^{\pi}(\ell/t)\sin\theta\, d\theta = 2\ell/(\pi t)$. Therefore $\widehat{\pi} = 2\ell/(t\widehat{p})$.

The page samples angle and center distance, counts crossings, and displays N, measured and expected crossings, and estimated versus true π.

Playground: darts and birthdays

Darts: a uniform point in the unit square lands in its quarter circle with probability area ratio $\pi/4$; estimate $\pi = 4h/N$.

Birthdays: no match for n people has probability $\prod_{i=0}^{n-1}(365 - i)/365$; match probability is one minus that product. At $n = 23$, it is 0.5073. The playground measures both against the exact model value.

Playground: collecting every coupon

With k types already held, each draw finds a new type with probability $(n-k)/n$. The expected wait for that next type is $n/(n-k)$.
Summing from $k=0$ to $n-1$ gives $E = n\sum_{j=1}^{n} 1/j = nH_n$. The playground compares mean draws across seeded runs with this expectation for $n = 10$ or 50.

Secretary rule: derive the finite-n chance

Set $r = \lfloor n/e \rfloor$. Skip the first r applicants and remember their best. If the overall best is at position $j > r$, the rule selects it exactly when the best of the first $j-1$ positions lies among the first r. That probability is $r/(j-1)$, while the overall best is at j with probability $1/n$. Therefore

$$P(\text{success}) = \sum_{j=r+1}^{n} \frac{r}{n(j-1)} \longrightarrow 1/e.$$

For $n = 100$, $r = 36$ and the finite sum is approximately 0.3710 (the limit is $1/e \approx 0.3679$).

What to remember

- A probability answer depends on the sample space and the host's rule.
- A reveal can change conditional probabilities; it does not erase the original choice.
- Derive the model first, then use simulation to check its numerical behavior.